



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO HARI AGENCY

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

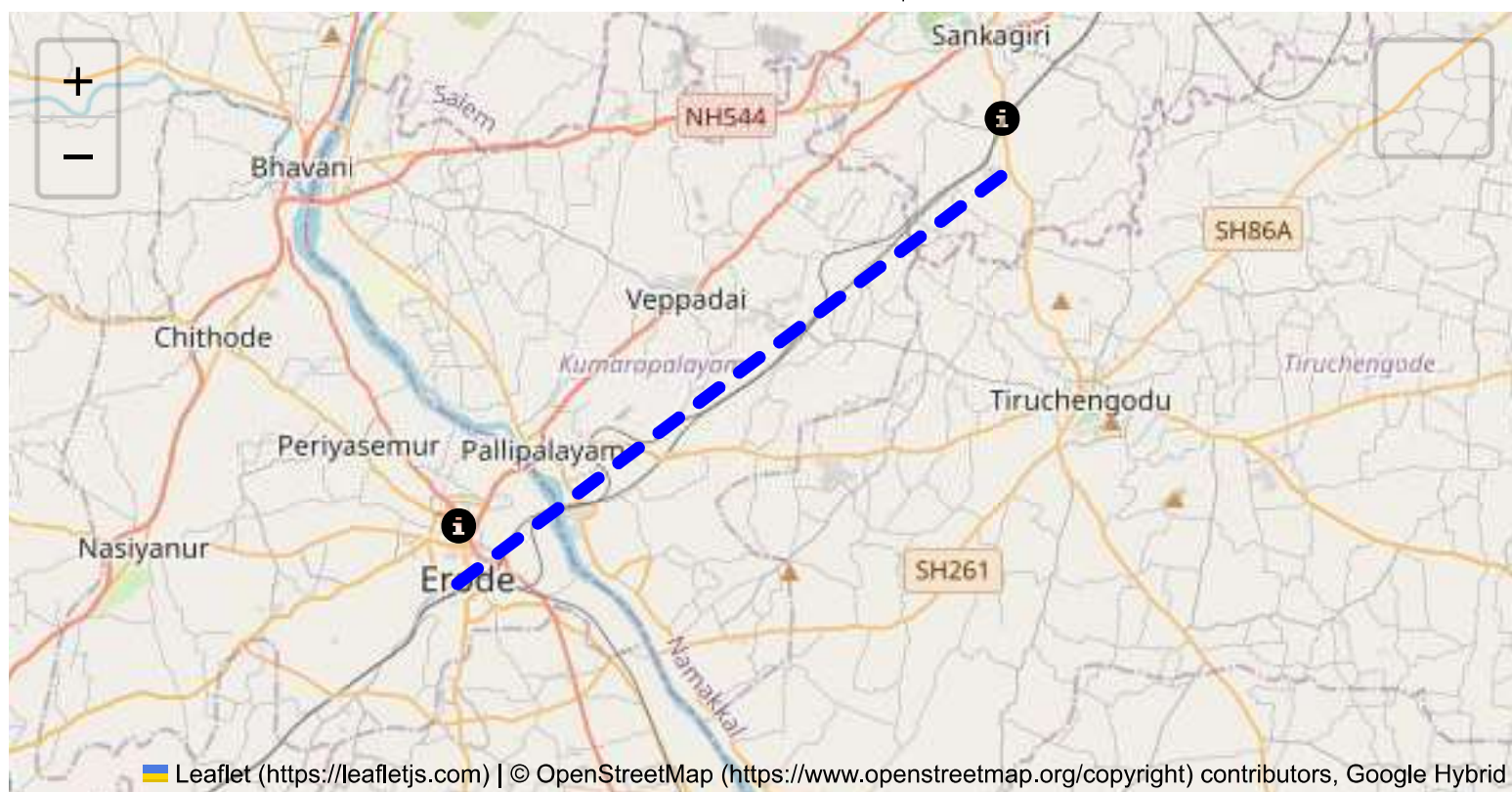
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:

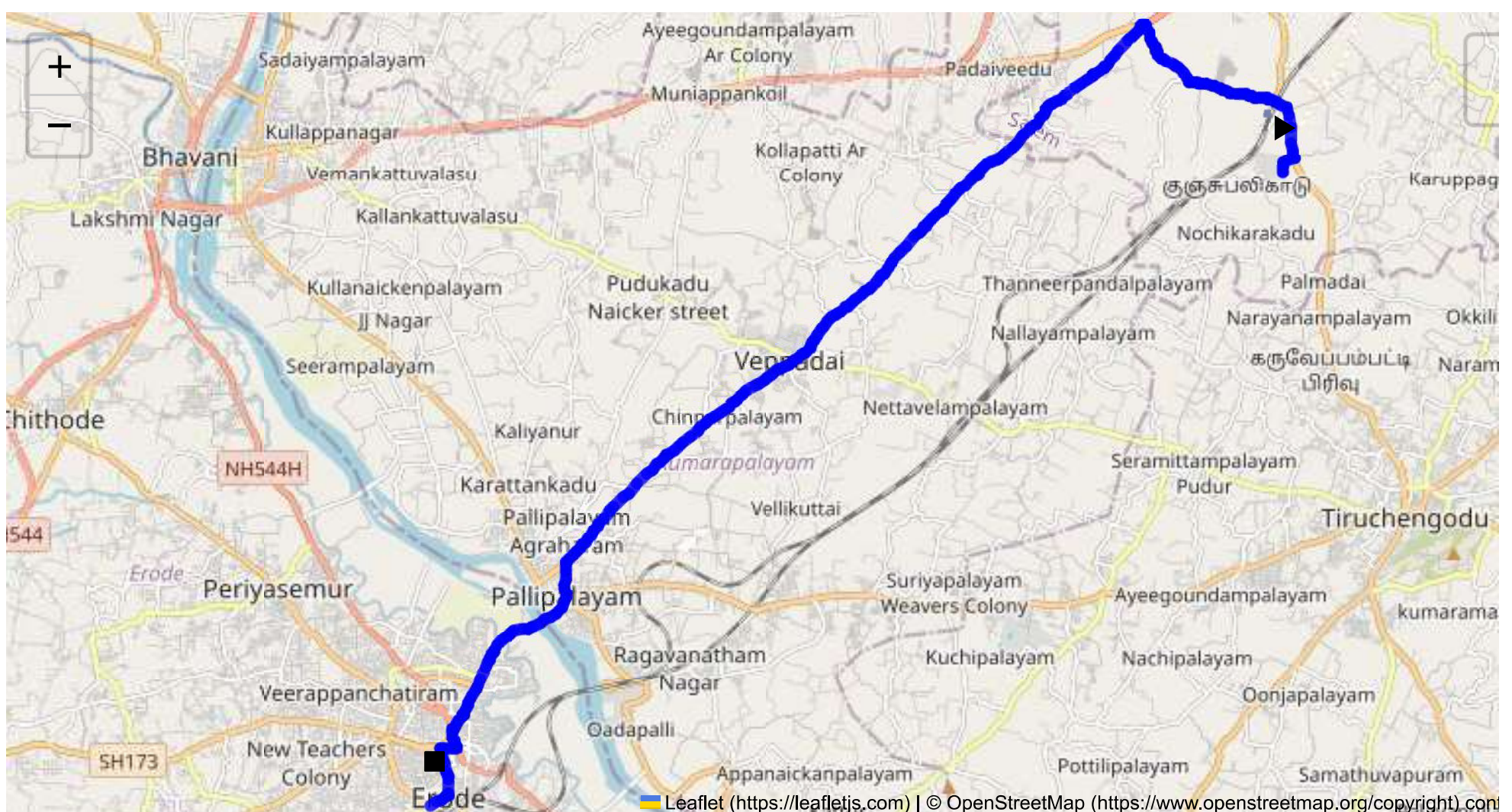
Total Distance: 26.09 km

Estimated Duration: 0.7 hours

Adjusted Duration (Heavy Vehicle): 0.9 hours

Start: (11.4381, 77.8734)

End: (11.32923, 77.724689)



Welcome to the Journey Risk Management Study

1. Overview of the Route Map

The route from Sangagiri to the Railway Station in Erode spans approximately 26.09 kilometers, primarily following major roads including NH544 and SH84. The journey typically takes around 43 minutes for heavy vehicles. The route mainly traverses urban and semi-urban areas, crossing through towns and agricultural zones before entering Erode.

2. Typical Weather Conditions and Potential Weather-Related Hazards

The region usually experiences a tropical climate with relatively high temperatures throughout the year. The monsoon season, from June to September, can bring heavy rains, leading to the risk of flooding and waterlogging on certain sections of the route, particularly in low-lying areas. During peak summer, extreme heat could affect vehicle performance, especially for those not well-maintained.

3. Analysis of Traffic Patterns

Traffic congestion is common during peak hours, typically from 8:00 am to 10:00 am and 5:00 pm to 8:00 pm, especially near major town centers and market areas. Erode's proximity to industrial regions can also contribute to heavy-duty vehicle traffic during these times. The entry and exit points of highways serve as major congestion zones.

4. Assessment of Road Quality and Infrastructure

The NH544 segment is generally well-maintained with smooth surfaces; however, some segments of SH84 may have uneven surfaces and require caution, especially for heavy vehicles. Watch out for road construction or maintenance activities, which could necessitate detours or slow down progress.

5. Suggestions for Alternative Routes for Emergencies

In case of emergencies, consider using the parallel service roads along NH544, which can provide quick escape routes. Using local roads to access nearby towns such as Salem and Perundurai may also be viable, ensuring GPS updates for any temporal changes.

6. Summary of Local Regulations Affecting Hazardous Material Transport

Tamil Nadu strictly regulates the movement of hazardous materials, requiring adherence to specific environmental regulations, use of approved containers, and clearly marked trucks. Permission from local authorities is necessary, particularly when traversing urban areas, due to traffic restrictions in residential zones.

7. Overview of Historical Incidents

The route has witnessed occasional incidents involving heavy vehicles, primarily due to overturns or brake failures on inclines. While incidents involving hazardous materials are rare, they underline the importance of strict adherence to safety protocols and maintenance schedules for vehicles.

8. Environmental Considerations and Sensitive Areas

Several segments of the route pass through agricultural lands, highlighting the need to prevent contamination from potential spills. Nearby waterways must be protected, requiring immediate action in case of leaks.

9. Analysis of Communication Coverage

For most of the route, mobile network coverage is strong; however, isolated stretches, particularly in less-populated areas, might experience weak signals or dead zones. Drivers should have emergency contacts available offline.

10. Estimated Emergency Response Times

Emergency response times can vary, typically ranging from 15 to 30 minutes within urban areas like Erode. In more remote sections, response times can extend to 45 minutes or longer, emphasizing the need for drivers to be prepared to manage emergencies initially on their own.

12. Overall Summary of Risk Assessment

The route from Sangagiri to Erode is generally safe for the transport of heavy vehicles carrying hazardous materials if all safety regulations and protocols are rigorously followed. Potential risks include weather-related issues during the monsoon, traffic congestion during peak hours, and road quality in specific segments. Regular training on emergency protocols, vehicle maintenance, and alternative routing would improve safety significantly.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.37086, 77.74828	15 KM/Hr
1	Turn	High	11.43811, 77.87348	15 KM/Hr
2	Turn	Medium	11.43956, 77.87340	30 KM/Hr
3	Turn	High	11.43968, 77.87345	15 KM/Hr
4	Turn	High	11.44029, 77.87544	15 KM/Hr
5	Turn	Medium	11.44898, 77.87410	30 KM/Hr
6	Turn	Medium	11.45348, 77.85706	30 KM/Hr
7	Turn	Medium	11.45352, 77.85698	30 KM/Hr
8	Turn	Medium	11.46318, 77.84913	30 KM/Hr
9	Turn	Medium	11.46312, 77.84886	30 KM/Hr
10	Turn	Medium	11.34493, 77.73032	30 KM/Hr
11	Turn	Medium	11.34483, 77.73022	30 KM/Hr
12	Turn	Medium	11.34470, 77.73018	30 KM/Hr
13	Blind Spot	Blind Spot	11.33920, 77.72902	10 KM/Hr
14	Turn	Medium	11.33914, 77.72659	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
15	Turn	High	11.33890, 77.72637	15 KM/Hr
16	Turn	Medium	11.33097, 77.72747	30 KM/Hr
17	Turn	Medium	11.33095, 77.72748	30 KM/Hr
18	Turn	High	11.33090, 77.72748	15 KM/Hr
19	Turn	Medium	11.33084, 77.72742	30 KM/Hr
20	Turn	Medium	11.33083, 77.72734	30 KM/Hr
21	Turn	High	11.33084, 77.72731	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Cheran Hospital	11.3680302, 77.7481976	30 km/h	Medium
2	pharmacy	Pharmacy	11.3652161, 77.7486375	30 km/h	Medium
3	hospital	Sri Kumaran E N T Hospital	11.3657814, 77.7477763	30 km/h	Medium
4	hospital	Sri Krishna Hospital	11.3656648, 77.7477783	30 km/h	Medium
5	hospital	Visaalam Hospitals	11.3651528, 77.747808	30 km/h	Medium
6	police	Police Station	11.3630177, 77.7476505	30 km/h	Medium
7	hospital	Government Hospital	11.3608508, 77.747453	30 km/h	Medium
9	hospital	Sri Hari Sakthivel Hospital	11.3526901, 77.7323055	30 km/h	Medium
10	hospital	Senthil Prakash Hospital	11.352639, 77.732267	30 km/h	Medium
11	hospital	V K Hospital	11.3513792, 77.7313976	30 km/h	Medium
12	hospital	Dhanvantri Critical Care Center	11.3504266, 77.7311274	30 km/h	Medium
13	hospital	Dhanvantri Critical Care Centre	11.35099, 77.730508	30 km/h	Medium
14	hospital	Indrajith Yoga Nature Cure Hospital	11.3499973, 77.732974	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
16	hospital	Kumudha Hospital	11.3467172, 77.7306797	30 km/h	Medium
17	hospital	City Hospital	11.3451151, 77.7300887	30 km/h	Medium
18	clinic	K.B.N. Nursing Home	11.344835, 77.7310771	30 km/h	Medium
19	clinic	Kpn Nursing Home	11.3437022, 77.7304525	30 km/h	Medium
20	hospital	Milan Hospital	11.343591, 77.733182	30 km/h	Medium
21	hospital	Kaviraj Ayurveda Hospital	11.3400349, 77.7298199	30 km/h	Medium
22	hospital	Dr Kokilasekar Nursing Hospital	11.3395874, 77.7269838	30 km/h	Medium
23	clinic	M.R Nursing Home	11.3386746, 77.7294262	30 km/h	Medium
25	hospital	Government Maternity Hospital, Agathiar	11.33716, 77.727075	30 km/h	Medium
26	clinic	Naseer Nursing Home	11.3364485, 77.7271691	30 km/h	Medium
27	hospital	Vikram Hospital, Erode	11.3356411, 77.7280569	30 km/h	Medium
28	hospital	Shabeer Hospital	11.33529, 77.7285236	30 km/h	Medium
29	hospital	J.B. Clinic And Hospital	11.3351769, 77.7282031	30 km/h	Medium
30	hospital	Dr. R.M. Habibullah Hospital	11.335246, 77.7266028	30 km/h	Medium
31	hospital	Kumarasamy Hospital	11.334273, 77.7259118	30 km/h	Medium
32	hospital	Thangam Hospital, Erode	11.3340961, 77.7262437	30 km/h	Medium
33	hospital	Ck Hospital	11.3326562, 77.7282011	30 km/h	Medium
34	hospital	Immick Hospital	11.332617, 77.728073	30 km/h	Medium
35	fire_station	Erode District Fire Station	11.3326366, 77.7272532	30 km/h	Medium
36	hospital	JJ hospital	11.33168, 77.728823	30 km/h	Medium
37	hospital	Jai Maaruthi Hospital	11.332, 77.727272	30 km/h	Medium
38	hospital	Shubiksha Hospital	11.3318174, 77.7299544	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
39	hospital	Bethesda Hospital	11.3314183, 77.7266231	30 km/h	Medium
40	hospital	Mercury Hospital	11.3328384, 77.7240531	30 km/h	Medium
41	hospital	Erode Ortho Centre	11.3294487, 77.7230172	30 km/h	Medium
42	hospital	Aditi Ent Hospital	11.3300862, 77.7236924	30 km/h	Medium
43	hospital	KRN Hospital	11.330054, 77.723136	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	Ssri Valliappa Vidhayalayam mat hr school	11.4004878, 77.7794381	30 km/h	Medium
8	marketplace	Weekly Market	11.3605669, 77.7488199	30 km/h	Medium
15	school	Municipality Girls Higher secondary school	11.3495618, 77.7325011	30 km/h	Medium
24	school	E.V.R. Govt Girls Hr. Sec School	11.3371265, 77.7266283	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High

Speed Limit: 15 KM/Hr
Coordinates: 11.37086, 77.74828



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44898, 77.87410



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45348, 77.85706



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46318, 77.84913



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.46312, 77.84886



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.34493, 77.73032



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.34483, 77.73022



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.34470, 77.73018



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.33920, 77.72902



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.33914, 77.72659



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.33890, 77.72637



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.33097, 77.72747



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.33095, 77.72748



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.33090, 77.72748



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.33084, 77.72742



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.33083, 77.72734



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.33084, 77.72731